

Lung Cancer Prediction

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Abstract

Lung cancer remains one of the deadliest cancers, and early detection is critical to improving survival rates. This study aims to develop a machine-learning model to assess the risk of developing lung cancer in people by analyzing factors such as age, gender, smoking habits, blood coughing, and chest pain. Identifying high-risk individuals can help medical professionals prioritize them for early diagnostic tests and timely treatment.

A dataset comprising 1,000 samples and 26 attributes was utilized. During data preprocessing, we addressed missing values and transformed the data into a format suitable for machine learning.

We employed multiple classification algorithms, including Random Forest, Decision Tree, and Logistic Regression, to predict lung cancer risk. The models were evaluated and compared using metrics such as precision, recall, accuracy, and F1 score to assess their predictive capabilities.

Our findings highlight the potential of machine learning models in enabling early identification of high-risk individuals, ultimately saving lives and reducing healthcare costs through timely intervention.

Introduction

Lung cancer is one of the most prevalent and life-threatening cancers globally, accounting for a significant proportion of cancer-related mortality. Despite advancements in medical science, the survival rate for lung cancer remains low due to delayed diagnosis and the aggressive nature of the disease. Early detection is critical for improving patient outcomes, yet traditional diagnostic methods such as imaging, biopsies, and manual assessments often have limitations. These include high costs, time delays, and variability in accuracy due to human interpretation.

In recent years, machine learning (ML) has emerged as a transformative approach to addressing these challenges in healthcare. As a subset of artificial intelligence, ML involves developing algorithms to learn patterns and relationships from complex datasets. In the context of lung cancer, ML models can analyze large volumes of medical data, such as radiographic images, genetic profiles, and clinical records, to provide predictive insights that are often beyond human capability. These models can potentially detect early signs of lung cancer, assess risk factors, and predict disease progression or treatment outcomes with remarkable precision and efficiency.

Integrating machine learning into lung cancer prediction not only enhances diagnostic accuracy but also supports personalized medicine by tailoring treatment plans to individual patients. This aligns with the broader goal of healthcare to move

from a reactive to a proactive model, where diseases are detected and managed at an earlier stage.

This project focuses on building a machine-learning model to predict lung cancer using patient-specific data. Key steps in this process include data preprocessing, feature selection, model training, and performance evaluation. By employing advanced techniques like cross-validation, the project aims to maximize the model's predictive accuracy and reliability.

Through this work, we aim to demonstrate the potential of machine learning to revolutionize lung cancer detection, enabling faster, cost-effective, and more accurate diagnostic solutions. This initiative represents a step forward in leveraging technology to improve clinical outcomes and enhance the quality of life for patients worldwide.

Related work

Model Selection:

Logistic Regression:

A statistical technique for binary classification tasks, such as yes/no, pass/fail, or disease/no disease, is logistic regression. The objective is to predict one of two potential outcomes based on one or more input variables.

Random Forest:

The Random Forest ensemble learning technique is mostly applied to tasks involving regression and classification. It creates several decision trees during training and aggregates their results to generate a prediction that is more reliable and accurate.

Decision Tree:

A decision tree is a machine-learning model that may be applied to tasks involving both regression and classification. It has a flowchart-like structure, with each internal node standing for a characteristic or attribute, each branch for a decision rule, and each leaf node for the label or result.

Data Visualization:

Name: Lung Cancer Patient Dataset

Samples: 980

Number of the Features: 12 Features Age, Gender, Air Pollution, Dust Allergy, Occupational Hazards, Smoking, Chest Pain, Coughing of Blood, Fatigue, Wheezing, Clubbing of fingernails, Frequent Cold.

Types: Numerical and Categorical.

Source: Kaggle

	Samples	Features
Before Preprocessing	1000	26
After Preprocessing	980	12

Table 1. Samples and Features Before and After Preprocessing

AGE: As depicted in Figure 1, the age ranges from 14 to 75. Notably, the maximum population is concentrated between 35 and 40 years

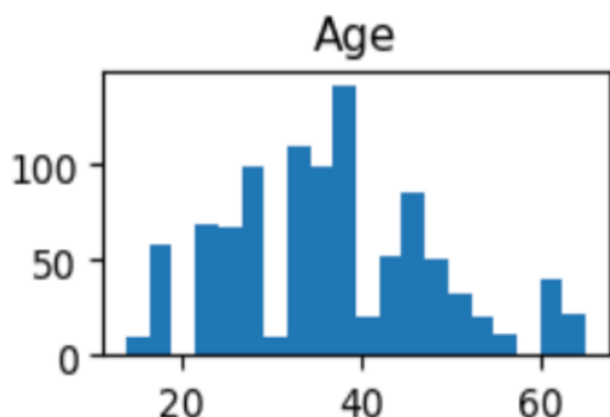


Fig. 1. Age Box plot

GENDER: From Fig 2, we can see that the count of males in this dataset is approximately 600, while the count of females is around 380.

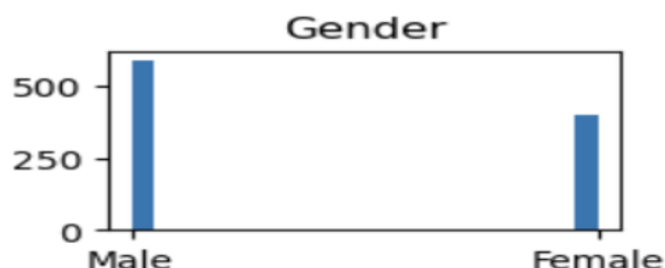


Fig. 2. Gender Box plot

Air Pollution: As shown in Fig 3, most of the data points lie near 6, indicating a significant prevalence of air-pollution-related health impacts.

Dust Allergy: As shown in Fig 3, most of the data points lie near 7, indicating a higher prevalence of dust allergy.

Alcohol Use: As shown in Fig 3, most of the data points lie near 2 and 8 indicating a higher prevalence of alcohol use.

Chronic lung disease: As shown in Fig 3 data point 6 has the highest prevalence of health impact.

Balance Diet: As shown in Fig 3 data points 2 and 8 have the highest prevalence of health effects.

Obesity: As shown in Fig 3 data point 7 has the highest impact.

Chest Pain: As shown in Fig 4, most of the data point 7.5 indicates a higher prevalence of chest pain-related health impacts.

Coughing of Blood: As shown in Fig 4, most of the data points near 3,4, and 7.5 indicate a significant prevalence of coughing of blood-related health impacts.

Fatigue: As shown in Fig 4, most of the data points 2.5 and 3, indicate a lower prevalence of Fatigue-related health impacts.

Wheezing: As shown in Fig 4, most of the data points 2,4, and 7 indicate a higher prevalence of wheezing-related health impacts.

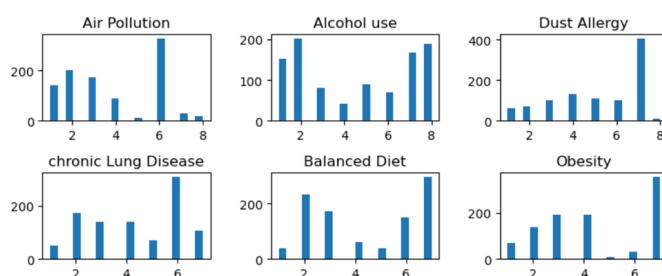


Fig. 3. Air Pollution Alcohol Use, Dust Allergy, Chronic Lung Disease, Balanced Diet, Obesity Box plot

Clubbing of Fingernails: As shown in Fig 4, most of the data points 2.5 and 4, indicate a higher prevalence of clubbing of fingernails related to health impact.

Snoring: As shown in Fig 4, most of the data point 2, indicates a higher prevalence of snoring-related health impact.

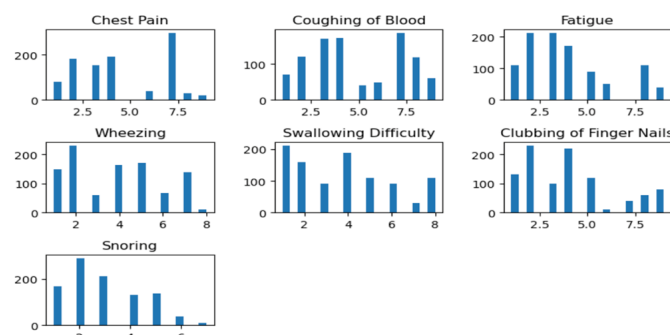


Fig. 4. Chest Pain, Coughing of Blood, Fatigue, Wheezing, Swallowing Difficulty, Clubbing of Finger Nails, Snoring

Occupational Hazards: As shown in Fig 5, data point 7, indicates a significant prevalence of occupational hazards.

Smoking: As shown in Fig 5, most of the data points lie near 1,2 and 7 indicating a lower prevalence of smoking-related health risks.

Frequent Cold: As shown in Fig 5, most of the data points 5 and 7, indicate a lower prevalence of frequent cold health risks.

Weight loss: As shown in Fig 5, most of the data points 2.5 and 6 indicate a higher prevalence of weight loss health risks.

Genetic risk: As shown in Fig 5, most of the data points 2 and 7 indicate a higher prevalence of smoking-related health risks.

Passive Smoker: As shown in Fig 5, most of the data points 2 and 7 indicate a higher prevalence of smoking-related health risks.

Shortness of breath: As shown in Fig 5, most of the data points 2 and 7 indicate a higher prevalence of smoking-related health risks.

Dry cough: As shown in Fig 5, most of the data points 2 and 7 indicate a higher prevalence of Dry cough health risks.

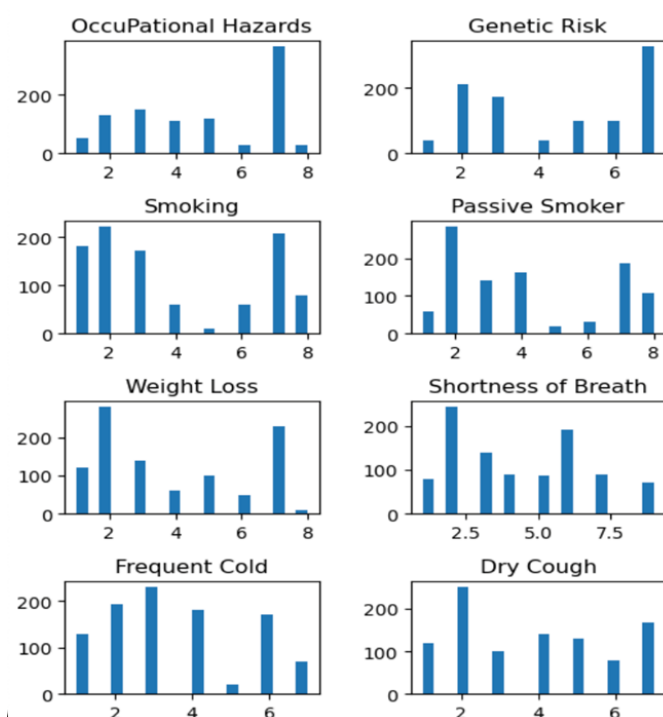


Fig. 5. Occupational Hazards, Genetic risk, Smoking, Passive Smoker, Weight Loss, Shortness of breath, Frequent cold, Dry cough Box plot

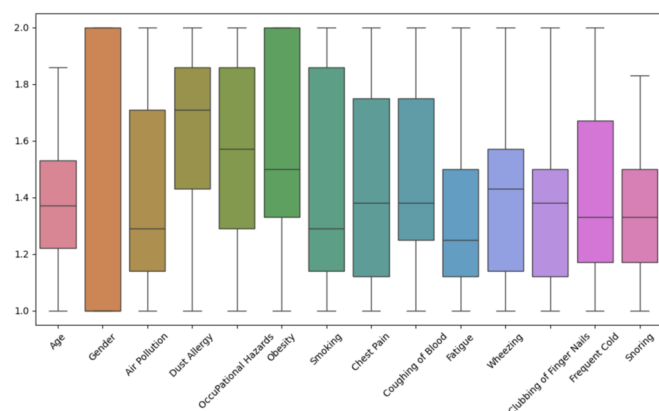


Fig. 6. Box Plot for all Features

This is a box plot visualizing the distribution of several variables across a dataset. Each variable is represented by a box-plot diagram, which provides insights into the statistical properties of the data for that variable.

Key Elements of the Box Plot:

1. Variables on the x-axis: The x-axis lists the variables under study, such as "Age," "Gender," "Air Pollution," "Dust Allergy," etc.

2. Values on the y-axis: The y-axis represents the scale of the data for each variable.

3.Box: The box spans the Inter quartile range (IQR), which is the range between the first quartile (Q1) and the third quartile (Q3).

4.Median: The horizontal line in the box represents the median value for that variable.

5.Whiskers: The whiskers extend to the smallest and largest data points that are not considered outliers.

Interpretation of Each Variable:

1.Age: The distribution is relatively narrow, with the median skewed toward the lower end of the scale.

2.Gender: It is binary, as the box does not represent a spread but distinct categories.

3.Right Skew (Positive Skewed): As depicted in Figure 6, the distribution of the given features exhibits a right skew. These features include Air Pollution, Obesity, Smoking, Chest Pain, Coughing of Blood, Fatigue, and Frequent Cold.

4.Left Skew (Negative Skewed): As depicted in Figure 6, the distribution of the given features exhibits a left skew. These features include Dust Allergy, Wheezing, and Clubbing of Finger Nails.

5.Normal Distribution: As depicted in Figure 6, the distribution of the given features exhibits a normal distribution. These features include Age, Occupational Hazards, and Snoring.

Methodology

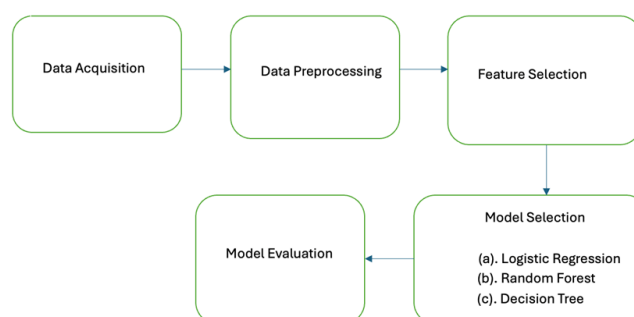


Fig. 7. Methodology

This flowchart illustrates the process of building and evaluating a machine-learning model. The first step is where data is gathered from Kaggle. The collected data involves cleaning and preparing the raw data to make it suitable for analysis. It includes tasks like handling missing values, removing duplicates, normalizing or scaling features, encoding categorical variables, and splitting the data into training and test sets. After preprocessing, the next step is identifying the most relevant features that contribute to the predictive power of the model. Feature selection helps in reducing dimensionality, eliminating noise, and improving model performance. Here different machine learning algorithms are chosen and applied to the prepared data. The flowchart mentions three algorithms Logistic Regression, Random Forest, and Decision Tree. Once the models are trained, their performance is assessed using evaluation metrics such as accuracy, precision, recall, and F1-score. This step determines which model is most effective for the task.

Step 1: Data Acquisition

In this step, we tried different datasets to find a balanced one.

Step 2: Data Preprocessing

There were no duplicates or null values in this dataset.

Data Understanding: Age is the only feature that has a numerical datatype, the rest of them are of categorical datatype. While the target class is an ordinal datatype.

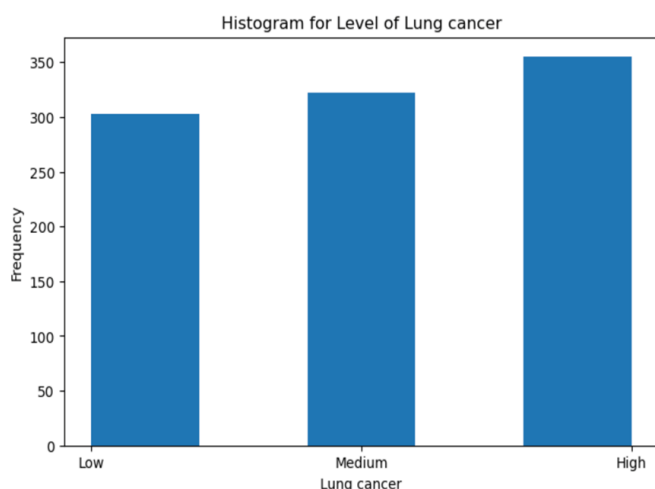


Fig. 8. This histogram represents a histogram for the levels of lung cancer, displaying the frequency of occurrences for three categories: "Low," "Medium," and "High."

Key Components of the Histogram:

X-Axis (Levels of Lung Cancer):

The categories: "Low," "Medium," and "High" represent the levels of lung cancer severity or risk.

Y-Axis (Frequency):

The frequency indicates the number of individuals or data points that fall into each category.

The height of each bar corresponds to the frequency for each level:

Low: Approximately 300 cases.

Medium: Approximately 320 cases.

High: Approximately 350 cases.

Distribution:

The "High" level has the most frequent occurrences, indicating that more individuals fall into the high-risk or high-severity category, and the "Medium" level follows closely behind. The "Low" level has the lowest frequency but still represents a significant number of cases.

Comparison:

There is a relatively balanced distribution across the three levels, with only slight differences in frequency.

It provides a clear overview of how lung cancer levels are distributed within the dataset. The relatively balanced distribution suggests that the dataset is comprehensive, covering all three categories nearly equally, which can aid in analyzing factors influencing lung cancer severity.

Fig 9 displays lung cancer results categorized by gender, with three levels of lung cancer risk: High, Low, and Medium.

Explanation of the Chart:

X-Axis (Gender): Divided into two categories, Female and Male.

Y-Axis (Count): Shows the number of individuals within each gender group who fall into each lung cancer risk category.

Color Legend:

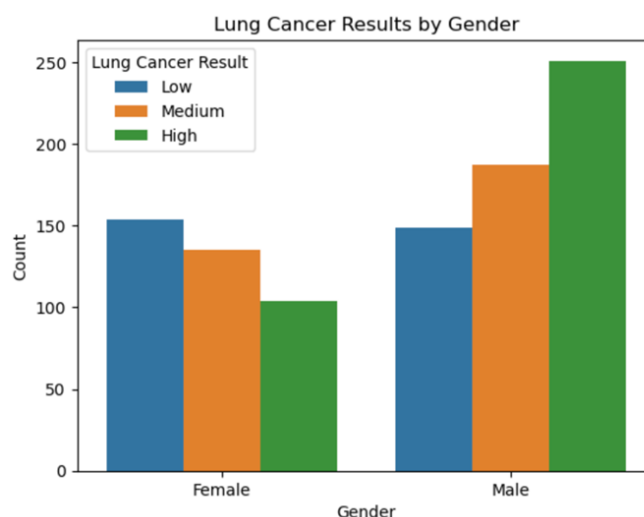


Fig. 9. Gender and Level of lung cancer

Blue - represents individuals with low lung cancer risk. **Orange** - represents individuals with medium lung cancer risk. **Green** - represents individuals with high lung cancer risk.

Observations:

For Females:

The Low-risk category has the highest count, indicating a larger number of females with a low lung cancer risk. The Medium risk count is slightly lower than the Low category. The High-risk count is the lowest among the three categories for females.

For Males:

The High-risk category has the highest count, showing more males are classified as high-risk. The Medium and Low-risk counts are similar but slightly lower than the High-risk. This chart highlights a potential difference in lung cancer risk distribution by gender, with males having a higher count in the High-risk category and females having a higher count in the Low-risk category.

Step 3: Feature Selection:

In this step, we selected features after implementing the important feature functions and confirmed by tree from the decision tree, as shown in Fig 10.

From Table 2, there are only 6 features which are important. So, we select those and add some more features like Age, Gender, Smoking, Frequent Cold, Fatigue, Dust Allergy, Chest Pain, and Occupational Hazards.

Step 4: Data Normalization:

It ensures that all features contribute equally to the model's learning process, preventing features with larger magnitudes from dominating those with smaller ones.

$$x = (x - \min(x)) / (\max(x) - \min(x))$$

Feature	Importance
Coughing of Blood	0.8635
Wheezing	0.7455
Chest Pain	0.7405
Dust Allergy	0.6667
Smoking	0.6136
Occupational Hazards	0.5766
Air Pollution	0.5745
Fatigue	0.5496
Clubbing of Finger Nails	0.4141
Age	0.4111
Frequent Cold	0.3643
Gender	0.0197

Table 2. Feature Importance Values

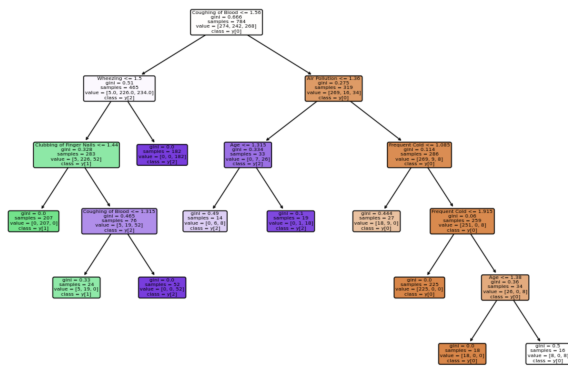


Fig. 10. Decision Tree

Measures of Assessment:

The following evaluation metrics will be used for classification tasks such as lung cancer prediction.

Accuracy: The ratio of correctly predicted instances to the total number of instances.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

Fig. 11. Accuracy

Precision: The ratio of true positives to the sum of true and false positives. This measures the accuracy of the positive predictions.

$$Precision = \frac{TP}{TP + FP}$$

Fig. 12. Precision

Recall: The ratio of true positives to the sum of true and false negatives. This measures how well the model can detect positive cases (i.e., individuals with lung cancer).

$$Recall = \frac{True\ Positive(TP)}{True\ Positive(TP) + False\ Negative(FN)}$$

Fig. 13. Recall

F1-Score: The harmonic means of precision and recall provide a balanced measure of the classifier's performance, especially in the case of imbalanced datasets.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Fig. 14. F1-Score

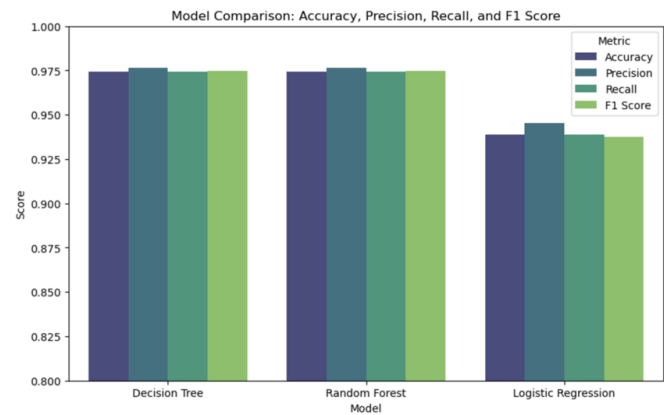


Fig. 15. Model Algorithms Comparison

Results and Discussion

1. Accuracy Comparison:

- Decision Tree: 97.5
- Random Forest: 97.5
- Logistic Regression: 93.88

2. Comparison:

Algorithm	Precision	Recall	F-1 Score
Decision Tree	0.98	0.97	0.97
Random Forest	0.98	0.97	0.97
Logistic Regression	0.95	0.94	0.94

Table 3. Algorithm Performance Metrics: Precision, Recall, and F1 Score

Algorithm	Accuracy	10-fold Cross-Validation
Decision Tree	0.97	0.987
Random Forest	0.97	0.991
Logistic Regression	0.93	0.918

Table 4. Algorithm Performance Metrics: Accuracy and Cross-Validation

Features	Decision Tree	Random Forest	Logistic Regression
Coughing of Blood, Wheezing, Fatigue	97%	98.5%	92.5%
Coughing of Blood, Smoking, Wheezing, Fatigue	96.5%	98.5%	97%
Dust Allergy, Wheezing, Fatigue	98%	99%	89%
Chest Pain, Occupational Hazards, Dust Allergy	98%	97%	87%

Table 5. Comparison of Models Based on Top Features

Conclusion

Random Forest is the best-performing algorithm in this study, with the highest cross-validation score (0.991) and strong metrics across all categories. Decision Tree performs almost as well as Random Forest, but its cross-validation score (0.987) is slightly lower, making it slightly less robust. Logistic Regression is outperformed by the tree-based models in all metrics, suggesting it may not be the optimal choice for this dataset. However, it could still be preferred for its interpretability and simplicity when computational efficiency is a priority. In summary, Random Forest is the most robust and accurate model for lung cancer prediction in this case. The visual analysis reveals that Age, chest pain, and coughing are the three most important factors in the early detection of lung cancer.

References

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